East West University
Department of Computer Science and Engineering
CSE400B Capstone Project

A Multimodal Approach for Developing a Robust Emotion Recognition System in Bengali

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1. Abstract

The limited availability of emotion recognition technology for Bangla hinders communication within Bangla-speaking communities. This research addresses this gap by introducing Anuvuti, a novel Bangla emotion recognition dataset specifically designed for real-time applications. Anuvuti facilitates the development of deep learning models for emotion detection by combining information from audio (speech) and visual (facial expressions) data, a technique known as multimodal fusion. We evaluated various models, including a custom CNN, LSTM, VGG16, and InceptionV3. The custom CNN achieved the highest accuracy (94.19%) on audio data, showcasing the effectiveness of multimodal fusion for Bangla emotion recognition. Anuvuti represents a significant advancement in this field, paving the way for more natural human-computer interaction in Bangla. Future work will focus on expanding Anuvuti for improved generalizability and incorporating a wider range of Bangla-specific emotions. A user-friendly web application for real-time emotion detection is also planned. Exploring transfer learning and incorporating text data are potential avenues for further improvement.Anuvuti's applications in education, customer service, and human-computer interaction systems hold promise to benefit Bangla-speaking communities by fostering more natural interactions. However, ethical considerations regarding user privacy, data security, and potential biases require careful attention throughout development. Overall, this research contributes to real-time Bangla emotion detection, promoting more natural human-computer interactions within Bangla communities.

Keywords: Bangla, Emotion Recognition, Multimodal Fusion, Deep Learning, Anuvuti

2.Introduction

Understanding emotions is crucial for effective human communication. Beyond words, both audio and visual cues play a significant role in conveying emotions. Accurately recognizing these emotions from audio-visual data holds immense potential in various fields, from healthcare and education to human-computer interaction.Traditional methods often focus on a single modality, such as facial expressions or vocal features. However, this neglects the richness of information available when analyzing both audio and visual data together. This research seeks to address this limitation by leveraging the power of deep learning for audio-visual emotion recognition in Bangla.Existing research in this area has explored the use of both audio and visual data for emotion recognition. However, existing datasets may not be suitable for specific languages like Bangla, or may lack the necessary diversity for robust deep learning models. Our research aims to overcome these limitations.The core challenge we address is developing a system that accurately recognizes emotions from Bangla audio-visual recordings. To achieve this, we create a novel dataset, ANUVUTI, specifically designed for Bangla emotion recognition. We then employ deep learning techniques to extract features from both the audio and video data, ultimately classifying the expressed emotions.This paper delves into the entire research process. We begin by outlining the limitations of single-modality approaches for Bangla data. We then introduce ANUVUTI, our custom dataset, and explain how it addresses the identified limitations. Following this, we detail the deep learning architectures used to extract features and classify emotions. Finally, we present the results, showcasing the effectiveness of our proposed approach compared to existing methods.Through this investigation, we contribute to the advancement of audio-visual emotion recognition using deep learning, specifically focusing on Bangla. We believe this research holds significant promise for various applications, paving the way for more nuanced human-computer

interaction and improved emotion analysis in diverse real-world scenarios focusing on the Bangla language.

3. Background

Automatic emotion recognition holds immense potential to revolutionize various applications like human-computer interaction (HCI), affective computing, and mental health support systems . However, achieving accurate and robust recognition remains a challenge due to the inherent complexities of human emotions and the limitations of single-modality approaches [1].Single-modality approaches, relying solely on speech or facial expressions, face inherent limitations. Firstly, they capture only specific aspects of emotions, leading to information loss and reduced accuracy, especially in real-world scenarios with diverse data and cultural influences [7, 8]. Secondly, these approaches often struggle to capture the intricate relationships between various features extracted from different data types, such as pitch, intensity in speech, and facial expressions [1]. While traditional processing methods using Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) achieve promising results, they might not effectively combine the spatial, temporal, and semantic information within each modality [1]. This can limit their ability to accurately recognize the full spectrum of emotions conveyed through speech or facial expressions.Furthermore, models trained on specific data distributions often perform poorly when encountering diverse data in real-world scenarios [3]. This lack of generalizability limits their practical applications, as they may not be able to adapt to the various ways individuals express emotions in different contexts. Additionally, relying solely on facial expressions can be susceptible to deception, as individuals can consciously control their outward expressions [6]. Moreover, focusing on a single modality might not capture the full spectrum of emotional cues, hindering accurate recognition.Finally, research on emotion recognition has primarily focused on resource-rich languages like English, leaving languages with limited resources like Urdu lagging behind [2]. This lack of publicly available emotional speech and facial expression databases hinders advancements in these languages and limits potential applications for these communities.To address these limitations, researchers are exploring multimodal emotion

recognition systems that combine information from multiple modalities, such as speech and facial expressions [7, 8]. This approach leverages the complementary nature of different modalities, providing richer and more comprehensive information for improved emotion recognition accuracy and robustness.Recent advancements in deep learning, particularly the development of powerful models like transformers, have opened up new possibilities for multimodal emotion recognition. Studies like [7, 8] demonstrate the effectiveness of combining features extracted from speech (using transformers) and facial expressions (using Action Unit extraction and Bi-LSTM models) to achieve significantly higher accuracy compared to individual modalities. This paves the way for developing more robust and effective emotion recognition systems in various real-world applications.However, several challenges remain in multimodal emotion recognition. One challenge is identifying and focusing on the most informative segments within each modality, particularly in video data where emotions might be dynamically expressed throughout a sequence [8]. Additionally, ensuring the generalizability and robustness of these systems across diverse data distributions and real-world scenarios requires further research and validation.Furthermore, studies like explore multimodal emotion recognition using neural networks on the IEMOCAP dataset, achieving promising results by combining information from speech, text, and motion capture data. This highlights the potential of combining various modalities for improved emotion recognition and provides valuable insights into different fusion techniques and approaches.We are excited by the potential of our Capstone project exploring multimodal emotion recognition for the Bangla language [13]. This innovative approach addresses the limitations of single-modality methods and leverages the complementary nature of speech and facial expressions to achieve more accurate and robust emotion recognition.Focusing on the Bangla language adds another layer of significance to the project. By tackling the underrepresentation of low-resource languages in emotion recognition research, we have the opportunity to contribute meaningfully to this field and pave the way for the development of applications that benefit Bangla speakers.

4. Related Work

Due to innovations on human emotion recognition has become a major concentration in the current century with the advancement of artificial intelligence techniques, there has been growing interest in multimodal emotion recognition in recent years. Although a lot of research has been done on this subject, minimal focus has been done on Bangla language. Researchers from all around the world use state-of-the-art methods, such as machine learning and deep learning, to recognize human emotions. We can observe the domain of the related studies into two main categories after reviewing and investigating various studies and works related to human emotions recognition:

4.1 Emotion classification and Recognition Techniques

Nafis Irtiza Tripto *et al.* proposed the underexplored area of sentiment analysis and emotion extraction in context of the Bangla language, which has received a comprehensive set of techniques based on deep learning models [1]. They included that model for both three class sentiment classifications (positive, negative, neutral, strongly positive, strongly negative) and the recognition of six basic emotions-anger, disgust, fear, joy, sadness and surprise. They achieved state-of-the-art results on a new dataset with a specific focus on analyzing YouTube comments in Bangla, English and Romanized Bangla, significantly outperforming baseline solutions by at least 10%. Their deep learning-based models such SVM and Naïve Bayes in sentiment and emotion classification task with setting a new benchmark in the field. However, the challenge of this paper is to develop new deep learning models that are specifically tailored to the emotion analysis in Bangla text.

Lutfiah Zahara *et al.* proposed the convolution neural network (CNN) method for feature extraction categorization of several facial expressions in real time using OpenCV library including security, human-computer interaction and video conferencing [2]. They using a facial picture dataset FER-2013 contains 35887 facial RGB images of different expressions and the 7 emotion that can be split down into the categories are: (angry, disgust, fear, happy, sad, surprise, neutral)

gathered from the google image such of each emotion and synonyms of the emotions. The limitations in this paper are that their accuracy (65.97%) is very low.

Akash Roy Choudhury *et al.* proposed Sequential Minimal Optimization (SMO), SVM and Random Forest machine learning method for classifications. They used 3 databases, these are EmoDB, SAVEES and TESS databases [3] . The SAVEES database contains 7 emotions (angry, disgust, sad, happy, fear, surprise, neutral) aged 27 to 31 years. The optimal average accuracy was achieved from TESS 'younger database, accuracy of SMO was 99.1% and accuracy of Random Forest was 99%.

M. ZAFAR IQBAL *et al.* Deep Convolutional Neural Network (DCNN), Bidirectional Long Short-Term Memory (BLSTM) and Time-Distributed Flatten (TDF)were unveiled a deep learning architecture for Speech Emotion Recognition (SER) layer to extract local sequential features from speech signal with developing a state-of-the-art SER system for both Bangla and English [4]. This study also included experimental results on cross-lingual SER using two datasets, including SUBESCO (taken from SUST Bangla Emotional Speech Corpus) and RABDESS which completely different languages and cultures and established that using aggregated datasets for model training greatly improves performance. The model achieved an accuracy of 78.6% on the SUBESCO dataset and 82.3% on the RAVDESS dataset with 7 emotions (angry, disgust, fear, happy, neutral, sad, surprise). The paper determined that the suggested design can be used to perform for SER in low-resource language and further research to explore more diverse datasets.

Linqin Cai *et al.* proposed that the paper used long short-term memory (LSTM) and convolution neural network (CNN) networks for voice emotion component and to extract facial expression data using small-scale kernel convolution blocks achieved accuracy 85.7% [5]. They IEMOCAP dataset which is widely used benchmark dataset for Malti-modal emotion identification, consists of data from speech, text, and motion capture data from face expression and hand movement, with 2 speaker per session includes 9 emotions (anger, happiness, excitement, sadness, frustration, fear, surprise, other and neutral state) with valence, arousal and dominance [6,7,25]. The knowledge gap on this paper in facial emotion recognition, the difficulty of small dataset and need of novel diagnostic techniques.

4.2 Bimodal emotion recognition on fusion

JUN-HWA *et al.* suggested a multi-modal approach to facial expression identification that uses a Modal Fusion Module (MFM) to combine audio and facial image data [6]. Both picture and audio data can have features extracted from them using the Swin Transformer. To resolve dataset imbalance, dynamic data resampling is also used. The model performs better on face expression identification tasks when tested in the Affective Behavior in-the-wild (ABAW) challenge. The fusion approach efficiently utilizes both modalities to improve facial emotion recognition.

Yongzhen Wu *et al.* presented a new method for identifying emotions by combining data from facial expressions and EEG signals. The approach uses a multi-level CNN model for facial expression and a stacked Bi-LSTM model for EEG signals, and combines the results at the decision level using the D-S evidence theory [7]. The experimental results show a significant improvement in emotion identification accuracy, with precision rates of 95.30% and 94.94% for valence and arousal dimensions respectively. The multi-modal fusion approach outperforms single-mode methods by 6.67% and 4.90% in valence and 7.62% and 5.10% in arousal dimensions, demonstrating the effectiveness of the proposed model in enhancing emotion identification accuracy.

Guichen Tang *et al.* proposed this research paper on recognizing emotions through the combination of audio and video data using an attention mechanism [8]. The study focuses on a feature fusion model that utilizes self-attention to effectively merge audio and video features. The results of the experiments show that the attention mechanism significantly improves recognition accuracy, with the proposed method outperforming state-of-the-art approaches on datasets such as eNTERFACE'05 and RAVDESS. The proposed multimodal emotion recognition model achieved an accuracy of 88.11% on the eNTERFACE'05 dataset and 93.23% on the RAVDESS dataset.

Dilnoza Mamieva *et al.* introduced a novel approach to emotion recognition by combining facial and speech features using an attention mechanism [9]. The fusion model integrates data from both modalities to enhance emotion detection accuracy. The study achieved high accuracy rates, surpassing 71% for each emotion class on datasets like CMU-MOSEI and IEMOCAP, demonstrating the effectiveness of the proposed fusion model in multimodal emotion detection.

Xusheng *et al.* proposed on human emotion recognition by optimally fusing facial expression and speech features. The proposed bimodal fusion algorithm combines facial expression recognition and speech emotion recognition using a weighted decision fusion method [10]. Experimental results demonstrate that the bimodal features-based emotion recognition model achieves better performance compared to uni-modal approaches. The accuracy of the proposed fusion model shows significant improvement in speech emotion recognition tasks, outperforming traditional algorithms.

SAMUEL KAKUBA *et al.* explored Deep Learning-Based Speech Emotion Recognition using a Multi-Level Fusion of Concurrent Features. It introduces the CoSTGA model, which leverages multi-level fusion of spatial, temporal, and semantic features to enhance the effectiveness and robustness of speech emotion recognition systems [11]. The study demonstrates that the multi-level fusion approach improves performance significantly compared to single-level fusion, with a notable increase in accuracy, precision, recall, F1 score, and area under the curve. The proposed CoSTGA model outperforms existing approaches by effectively integrating spatial, temporal, and semantic feature representations, showcasing its potential for advancing speech emotion recognition technology.

Shiqing Zhang1 *et al.* proposed audio-visual emotion recognition by integrating facial expression and affective speech features [12]. The study uses local binary patterns for facial image representations and acoustic features for speech emotion recognition. By fusing these modalities at the feature-level and employing support vector machines for classification, the proposed method achieves an accuracy of 66.51% in emotion recognition, outperforming single modality approaches. The fusion model demonstrates the effectiveness of combining facial expression and affective speech for improved audio-visual emotion recognition.

4.3 Overview of The Related Work Section

Based on an extensive review of the academic literature and studies, this related work summary table provides an overview of the most related research and methodologies in the area of human emotion recognition in both facial and speech with the use of deep learning and machine learning technologies.

Table I: Summary of literature review in Emotion Detection from Audio and Facial Expression.

Author(s)	Methods	Accuracy / Results	Contributions	Limitations
Nafis Irtiza Tripto *et al.* [1]	LSTM, CNN, SVM	LSTM - 65.97% SVM- 54.34%	Comprehensive set of techniques to identify sentiment and extract emotions from Bangla texts using deep learning models	Only used Bangla text
Lutfiah Zahara *et al.* [2]	CNN	65.97%	Design and implementation of a system that can predict and recognize the classification of facial emotions in real-time based on feature extraction	Low accuracy

Akash Roy Choudhury *et al.* [3]	SMO, SVM, RF	SMO=99.1%, RF= 99%	Combining excitation source features with spectral features for emotion recognition from speech signals	Lack of normalization performed on the speech signals before feature extraction and classification
M. ZAFAR IQBAL *et al.* [4]	DCNN, SVM, BLSTM, LSTM	CNN= for the SUBESCO dataset 86.9% and 82.7% for RAVDESS datasets	A novel architecture consists of a feature learning block DCNN, a time-distributed flattening layer, and a BLSTM layer, enabling the model to capture both local and sequential information from emotional speech signals	Lack of investigation into cross-lingual transfer learning factors

Linqin Cai *et al.* [5]	CNN, V-LSTM, DNN	85.2%	By integrating CNN and LSTM networks for speech emotion feature extraction and designing multiple small-scale kernel convolution blocks for facial expression feature extraction	Issues with lighting, pose and occlusion variations
JUN-HWA *et al.* [6]	Modal Fusion Module (MFM)	-	a novel approach to facial expression recognition by fusing audio and visual modalities using a Modal Fusion Module and Swin Transformer, addressing	lack of detailed analysis or comparison with other state-of-the-art models

			dataset imbalances through dynamic sampling	
Yongzhen Wu *et al.* [7]	multi-level CNN model, Bi-LSTM model	CNN =96.36% (CK+ dataset) and 73.00% (FER2013 dataset). Fusion =95.30% and 94.94% respectively (dimensions of valence and arousal)	combines facial expression image information with EEG data	low usability and scalability of the proposed approach
Guichen Tang *et al* [8]	VGG16	88.11%	improving the model recognition performance, leading to better results on datasets	potential issue of overfitting in deep networks
Dilnoza Mamieva *et al.* [9]	CNN, MFCC, ResNet, LFGP AND HFGP for feature extraction	CNN= IMOCAP (WA of 74.6% and an F1 score of 66.1%), CMU-MOSEI (WA of 80.7% and F1 score of 73.7%)	effectively capture and integrate relevant information from both facial and	potential vulnerability introduced by the attention mechanism.

			speech modalities	
Xusheng *et al.* [10]	CNN, RNN, MFCC, LSTM	Facial expression recognition Speech emotion recognition RML (Facial expression recognition 0.6, Speech emotion 0.4) AFEW (Facial expression recognition=0.75 ,Speech emotion recognition=0.25) eNTERFACE'05 (Facial expression recognition=0.8, Speech emotion recognition= 0.2)	The developed bimodal fusion approach can be applied in real-time scenarios to analyze and interpret human emotions.	noisy data and limited sample sizes
SAMUEL KAKUBA *et al.* [11]	BiLSTM, CoSTGA, feature extraction GFLB and LFLB	After fusion CoSTGA for weighted accuracy 75.82% and unweighted accuracy 75.50%	The concurrent learning of spatial, temporal, and semantic features at different fusion levels improves the effectiveness	its reliance on the dyadic database for evaluation, which may not fully capture the complexity and variability present in real-

			and robustness of speech emotion recognition models.	world speech scenarios.
Shiqing Zhang1 *et al.* [12]	Mel-Frequency Cepstral Coefficients (MFCC), support vector machines (SVM), local binary patterns (LBP), Modality Fusion	Facial expression: 62.79% Affective speech: 44.73% After fusion (Audio-visual): 66.51%	demonstrating the effectiveness of integrating facial expression and affective speech for improved audio-visual emotion recognition, achieving a 66.51% accuracy in emotion classification.	the recognition accuracy for some emotions, such as disgust, fear, and surprise, was relatively low, with accuracies below 61%

5. Problem Statement

Human emotions play a crucial role in communication and social interaction. Automatic emotion recognition holds immense potential to revolutionize human-computer interaction (HCI) by enabling computers to understand and respond to the emotional state of users [1]. However, achieving accurate and robust recognition, especially for languages with limited resources like Bengali, remains a significant challenge [1]. The current focus on resource-rich languages in emotion recognition research leaves Bengali speakers without the ability to leverage this technology for more natural and effective HCI experiences.

Existing emotion recognition approaches often rely on a single modality, such as speech or facial expressions [1, 3, 4]. These methods suffer from inherent limitations. Speech-based recognition can be influenced by cultural variations in speech patterns and the ability to mask emotions through vocal control [3]. Facial expressions alone might not capture the full spectrum of emotional cues and can be susceptible to misinterpretation due to cultural differences [3]. Additionally, single-modality approaches fail to capitalize on the richer information available by combining different modalities [1].

Recent advancements in multimodal emotion recognition offer a promising solution by fusing information from multiple sources, such as speech and facial expressions [14, 15]. This approach leverages the complementary nature of different modalities, providing a more comprehensive picture of an individual's emotional state [14, 15]. Studies have shown that multimodal fusion can significantly improve emotion recognition accuracy compared to single-modality approaches [6, 7, 8, 14, 15].

However, a critical gap exists in research on emotion recognition for languages like Bengali. The lack of publicly available emotional speech and facial expression databases specifically for Bengali hinders the development of robust emotion recognition systems for this language [1]. Existing systems trained on English data might not generalize well to Bengali due to potential cultural differences in emotional expression [3].

Furthermore, while multimodal emotion recognition has shown promise, there is a continuing need for research that explores the development of real-time emotion detection systems that can function effectively in dynamic environments [14, 15]

Our project, Anuvuti, aims to bridge this gap by developing a robust and generalizable real-time emotion detection system specifically designed for Bengali. We will leverage the Bengali emotional speech and facial expression dataset we have meticulously created. This valuable resource will serve as the foundation for training our real-time emotion detection system. Secondly, inspired by recent advancements in multi-level fusion and concurrent feature learning, Anuvuti will employ multimodal fusion with deep learning. This approach entails combining information from both audio (speech) and image (facial expressions) data within our Bengali dataset. By harnessing this richer set of emotional cues, Anuvuti aims to achieve robust real-time emotion recognition in Bengali. Finally, by applying deep learning techniques, we will develop a system capable of analyzing audio and visual data in real-time. This will enable Anuvuti to understand the emotional state of Bengali users, fostering more natural and user-friendly interactions between Bengali speakers and computers and applications.

We believe that our research will make a significant contribution to the field of emotion detection in the Bengali language. By achieving these goals a robust and generalizable emotion detection model that can be used to improve the lives of millions of people around the world.

6. Research Question

This research addresses this gap by introducing Anuvuti, a real-time emotion detection system specifically designed for the Bengali language. We propose leveraging the power of multimodal

fusion with deep learning to overcome the limitations of single-modality approaches. Our meticulously created Bengali emotional speech and facial expression dataset will serve as the foundation for training Anubhuti.Considering these queries we are working to make an effective project after which we will get the answer of :

RQ1. Multimodal Fusion for Robust Bangla Emotion Recognition: How can deep learning models be effectively utilized for multimodal fusion (audio and visual data) to achieve robust and generalizable real-time emotion recognition in Bangla?

1.1: How can we effectively combine features extracted from audio and visual data to create a robust and informative representation for emotion recognition in the Bangla language?

1.2: Which multimodal fusion techniques (e.g., early fusion, late fusion) are best suited for combining audio and visual features within deep learning models for real-time Bangla emotion recognition?

RQ2.Real-time Bangla Emotion Detection Web Application: How can we develop a user-friendly web application for real-time Bangla emotion detection using the insights from multimodal fusion?

2.1: What design principles should be followed to create a user-friendly web application for real-time Bangla emotion detection, considering factors like ease of use, accessibility, and cultural appropriateness?

7. Objective:

Anuvuti, our real-time emotion detection system designed for Bangla, leverages a meticulously created Bengali emotional speech and facial expression dataset for training. Our aim is to bridge

the gap in emotion recognition technology for low-resource languages by developing a robust and generalizable real-time Bangla emotion detection web application. This application will achieve this by harnessing the power of multimodal fusion, combining information from both audio (speech) and visual (facial expressions) data using deep learning.

The following research objectives outline the next phase of Anuvuti's development:

- **Evaluate Anuvuti's Performance on Multimodal Emotion Recognition**
    - To assess the impact of multimodal fusion on emotion recognition accuracy. We will directly influence the web application's effectiveness in combining audio (speech) and visual (facial expressions) data for real-time emotion detection.
    - To investigate emotion-specific performance of Anuvuti's multimodal approach in recognizing different emotions (happy, sad, angry, etc.) to identify potential strengths and weaknesses.
- **Optimize Anuvuti for Real-time Processing:**
    - To investigate techniques for optimizing Anuvuti's processing speed to achieve real-time emotion recognition within the web application. This is crucial for ensuring smooth user interaction.
    - To evaluate the trade-off between accuracy and processing speed to find the optimal balance for the web application. Maintaining accuracy while achieving real-time functionality is essential.
- **Develop and Test a Real-time Bangla Emotion Detection Web Application**
    - To design and develop a user-friendly web application that integrates Anuvuti's real-time, multimodal emotion recognition capabilities for Bangla. This application should provide users with immediate feedback on their emotional state during interaction, leveraging both audio and visual data streams.
    - To test and evaluate the web application's functionality and user experience with real users. This involves assessing factors like accuracy, ease of use, and overall user satisfaction. Our goal is to refine the application for a positive user experience that effectively detects emotions in real-time using multimodal fusion.

By systematically addressing these objectives, we can establish Anuvuti as a leading real-time

emotion detection system that excels at multimodal fusion for Bangla. This paves the way for our web application to analyze real-time Bangla user emotions through a comprehensive understanding of both speech and facial expressions.

8. Methodology

We used some widespread deep learning models in our project. The models are trained to detect multi-emotion using audio and facial expression images. First, we built our custom video dataset,

collecting seven emotional expressions from 10 participants. Then, we extracted an audio file and an average of 20 facial frames from each video. We stored those audio files in an audio dataset and facial frames in an image dataset. After that, we perform some pre-processing steps separately on audio and image datasets. From audios, we extracted MFCC (Mel-Frequency Cepstral Coefficients) which is a feature widely used in speech and audio processing to represent the short-term power spectrum of a sound. Then we performed some preprocessing on the extracted MFCC features. We performed noise addition, time stretching, and pitch shifting, data augmentation as pre-processing. In images, on average twenty 128 x 128 pixels facial frames were extracted from a video and saved in JPG format. After that, normalization, rescaling, resizing, noise reduction, extracted AUs, and grayscaling were performed as pre-processing steps on them. After following the necessary pre-processing, we split the two datasets into train and test datasets and fitted the training dataset into the model. Once the image and audio test datasets were trained by the deep learning models, then we fitted those models into test datasets. We performed and conducted accuracy, precision, validation, loss functions, and confusion metrics of our Models to assess how well the models are performing.

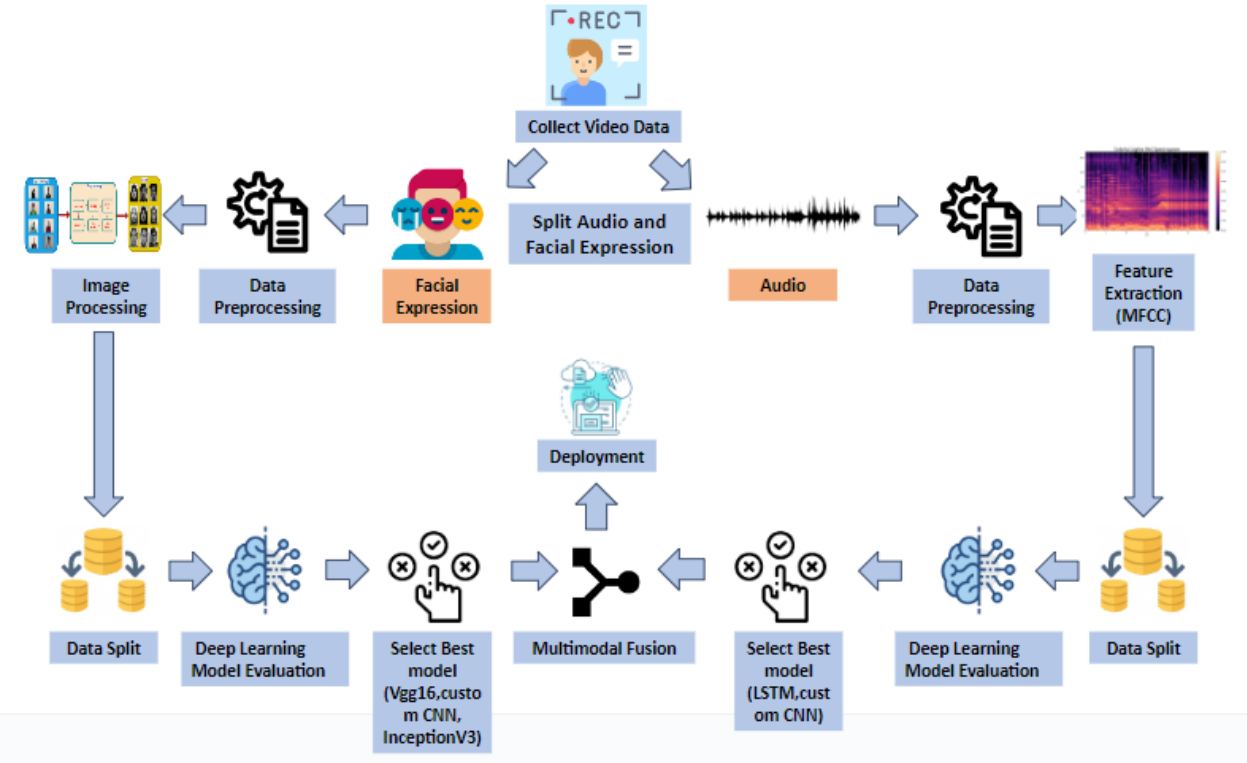

Fig 1: Flow-chart of The System Design.

Traditional Machine learning models excel with comparatively smaller datasets. Because of the large amount of image and audio data in our datasets, deep learning technologies were used and provided us with significant advantages. Deep learning excels at automatically extracting complicated features from large datasets. Unlike traditional methods that require hand-crafted feature engineering, deep learning models may learn these features directly from data, potentially leading to better performance in recognizing emotions from facial expressions and voices. This is especially useful for collecting the small subtleties in face features and temporal shifts in audio data, which are critical for reliable emotion identification.

8.1.Dataset

By examining his/her video stream, the audio-visual emotion detection system determines the emotional state of the speaker. Over the past 20 years, academics' attention has been increasingly focused on Speech Emotion Recognition (SER) research. It is one of the most difficult fields of research implementing deep machine learning and artificial intelligence frameworks, and it is a crucial yet challenging assignment in human-computer interaction systems. Though Bangla is the seventh most spoken language in the world, due to Insufficient data for audio-visual emotion identification tasks it is regarded as a low-resource language.

Collection of raw video samples:

It appears that there are no widely audio-visual emotion recognition datasets available in Bangla on this type of research. So, we mainly decided on generating our own dataset working on both audio and facial emotion recognition systems, along with the sets of seven emotions (Happy, Sad, Anger, Fear, Surprise, Neutral and Disgust) in Bangla. Here we are introducing ANUVUTI is an audio-visual emotion recognition dataset to deal with this issue. This dataset consists of video recording from 10 participants where 6 non-professional actors are participating (age group between 18 and 25 years) in seven emotions. Video recordings were collected in five phases with 10 participants representing seven human emotional states—angry, disgust, fear, neutral, sad, surprise and happy—are included in this dataset for 35 sentences (we used five sentences for each emotional state) in a realistic sense. The total number of recording involves 5 sentences × 3 repetitions × 6 emotional states (angry, happy, sad, fear, disgust and surprise) × 10 participating speakers = 900 video recordings + 5 sentences × 3 repetitions × 1 emotional state (neutral) × 9

participating speakers = 135 recordings, resulting in a total number of recordings of 1032. The recordings' durations vary from 1 to 4 seconds and are not fixed. This duration is determined by the speaker's time and this length depends on the sentence structures. So, the total duration of the video recording is 35 min. This dataset is collected by recording through smart phone and laptops. We used MiniTool MovieMaker software to trim video recordings with a vertical resolution of 720p (pixels) and video's frame rate 25fps.

Split speech audio recordings from video:

We used code to extract audio from video using the moviepy.editor module in Python. It loads each video, extracts the audio, and saves it as a WAV file while iterating over the video files in a designated folder. The write_audiofile method is used to save the extracted audio, while the VideoFileClip class is used to load videos and extract audio. The audio file contains 44.1 kHz default sample rate. Extracted audio was stored in seven classes (balanced). This contains 1035 samples of different emotions. All the actors were assigned a sequence number when they were chosen, and the sentences have been methodically documented in a prearranged sequential order.

The file name of the audio dataset's structure: P_<person-number><1st-emotion><sentence-number>_(<num-of-repeat).

Split image from video:

The OpenCV library is used by Python code to extract frames from video files. It takes a defined number of frames (by default, 28) out of the center of each video clip in a given folder and saves them as a .jpg file. It iterates through each video file in a specified folder, extracts frames, and renames (Emotion's name_sequential number_face) them. This dataset consists of 1280 × 720 dimension (width 1280, height 720) with (6 emotional states (angry, happy, sad, fear, disgust and surprise) × 5 sentences× 3 repetitions × 20 frames of the center of each video × 10 participating speakers + 1 emotional state (neutral) ×2,157 = 18,000 + 2,157 RGB images = 20,157 RGB images

with each image labeled with one or more of seven basic emotions: angry, disgust, fear, happy, sad, surprise, and neutral.

Specifications of ANUVUTI:

The most challenging task of SER is a language-specific field of study and finding an appropriate dataset with distinctive characteristics of the target emotions. The developed ANUVUTI dataset is collected in real time. Finally, a descriptive summary of the ANUVUTI dataset is summarized in Table 2 for a better understanding of the dataset:

Table 2: descriptive summary of the ANUVUTI dataset:

Summary	Descriptions
Used language	Standard Bangla
Dataset type	Realistic
File type	Audio-visual
File format of video clips	Mp4
File format of audio clips	WAV
Sampling rate (video clip)	44.1 kHz
Sampling rate (audio clip)	44.1 kHz
Emotion states	Neutral, Happy, Sad, Angry, Disgust, Fear, Angry
Number of emotions	7
Number of actors	10 (6 non-professional female and 4 non-professional male)
Total number of video clips	1032

Number of speech dialogs	1032
Audio duration per clip	1-4 sec
Total duration of the dataset	2070 Sec
Size of the dataset	497 MB
Utilized software	MiniTool MovieMaker
Speakers Age Group	18 -25 years
Number of frames	20,157
Number of Sentences	35 (each emotion contains 5 sentences)
Video Source	Real time

Description of Sentences:

There are 35 sentences that are used to prepare the ANUVUTI dataset. They are:

Happy:

১.বাংলা: আজকে আমি অনেক খুশি।

English: I am very happy today.

phonemic: ajke ami onek khusi

২.তোমার সাথে দেখা হওয়াতে অনেক ভালো লাগছে।

English:Nice to meet you.

phonemic: tOmar sathe dekha hoOyate onek valo lagche.

৩.আজ আমার চাকুরী হয়েছে।

English:I got my job today.

phonemic: aj amar cakuri hoyeche.

৪.আমি পরিক্ষায় খুব ভালো করেছি।

English:I did very well in the exam.

phonemic: ami porikkhay khub valo korechi.

৫.বাহ! আমি তোমার সফলতায় অনেক খুশি হয়েছি।

English:Wow! I am very happy with your success.

phonemic: bah! Ami tOmar sofolotay onek khushi hoyechi.

Sad:

১.তার মৃত্যু আমি কল্পনাও করতে পারি নি।

English:I could not imagine his death.

phonemic:tar mrritzu ami kolponaoO korte pari ni.

২.ইস! দূর্ঘটনাটি খুবই মর্মান্তিক।

English:Yes! The accident is very tragic.

phonemic: ishh! dUrrghTonaTi khuboi morrmantik.

৩.তোমার সাথে এটি খুবই খারাপ হয়েছে।

English:It's too bad with you.

phonemic: tOmar sathe eti khubi kharap hoyeche.

৪.আজকের দিনটি আমার জন্য ভালো যায় নি।

English:Today has not gone well for me.

phonemic:Ajker dinti amar jonno valo jay ni.

৫.কখনও কখনও ভাগ্য আমাদের পক্ষে থাকে না।

English:Sometimes luck is not on our side.

phonemic: Kokhono kokhono vaggo amder pokkhe thake na.

Angry:

১.আমার সাথে একদম চিল্লাবা না।

English:Don't mess with me at all.

phonemic:Amar sathe ekdom cillaba na.

২.তুমি এই ব্যাপারে আমার থেকে বেশি জানো?

English:You know more about this than me?

phonemic: tumi ei bzapare amar theke besi jano?

৩.এখান থেকে যাও তুমি।

English:You go from here.

phonemic:ekhan theke zau tumi.

৪.এত বিরক্ত করবা না তো।

English:Don't be so annoying.

phonemic:Eto birokto korba na tO.

৫.মুখ সামলে কথা বলবে।

English:Speak with mouth control.

phonemic: Mukh samle kotha bolbe.

Disgust:

১.উফ! কী বাজে দূর্গন্ধ!

English:Oops! What a bad smell!

phonemic:uff! Ki baje durgondho!

২.উফ! তুমি এই কাজটি কেন করলে?

English:Oops! Why did you do this?

phonemic: Uff! tumi ei kajti keno korle?

৩.আপনি আমার দিকে এভাবে তাকিয়ে আছেন কেন?

English:Why are you looking at me like this?

phonemic:Aponi amar dike evabe takiye achen keno?

৪.উফ! সে খুবই বিরক্তিকর।

English:Whoops! He is very annoying.

phonemic: Uff! Se khubi biroktikor

৫.উফ! তাকে দেখলেই আমার গা জ্বালা করে।

English:Whoops! Just seeing him makes my heart burn.

phonemic:Uff! take dekhlei amar ga jala kore.

Fear:

১.আমার অনেক ভয় লাগছে।

English:I feel very scared.

phonemic: Amar onek voy lagche.

২.কে! কে! কে! ওখানে।

English:Who! Who! Who! there.

phonemic: ke!ke!ke! Okhane

৩.আগামীকালের পরীক্ষা নিয়ে আমি খুবই ভয়ে আছি।

English:I am very scared about tomorrow's exam.

phonemic:Agamikaler porikkha niye ami khuboi voy e achi.

৪.সাপটি আমার দিকে আসছে!

English:The snake is coming towards me!

phonemic:Sapti amar dike asche.

৫.হরর মুভিটি খুবই ভয়ংকর ছিল।

English:The horror movie was very scary.

phonemic:Horor muviti khuboi vayongkor chilo**.**

Surprise:

১.আপনি ফিরে এসেছেন!

English: You are back!

phonemic: Apnoi fire esechen!

২.উপহারটি আমার জন্য!

English:The gift is for me!

phonemic: upoharti amar jonno?

৩.বাহ! কী সুন্দর একটি জায়গা।

English:Wow! What a beautiful place.

phonemic:Bah!ki sundor ekti jayga!

৪kor তার আচরনে অবাক হয়েছি।

English:I am surprised by his behavior.

phonemic:Ami tar achorone ovak hoyechi

৫.বাহ! তুমি এই কাজটি এত তাড়াতাড়ি কিভাবে করলে?

English:Wow! How did you do this so quickly?

phonemic:Bah! tumi ei kajti eto taratari kivabe korle?

Neutral:

১.আমি ভালো আছি।

English:I am fine.

phonemic:Ami valo achi.

২.আমি আজকে যেতে পারব না।

English:I can't go today.

phonemic:Ami ajke zete parbo ń'a.

৩.আপনি এই কাজটি করতে পারবেন?

English:Can you do this?

phonemic:Apnoi ei kajti korte parben?

৪.আমি বিকালে বের হবো।

English:I will go out in the afternoon.

phonemic:Ami bikale ber hobo.

৫.আমি গল্পের বই পড়তে পছন্দ করি।

English:I like to read story books.

phonemic:Ami golper boi porte pochondho kori

Emotion recognition Datasets in Bangla Language:

Table 3: A comparison Among Various Public Bangla Language ER and ANUVUTI Dataset

Specification	SUBESCO	BanglaSER	BANSp Emo	KBES	ANUVUTI
Number of records (files)	7000	1467	792	900	**1035**
Number of Emotions	7	5	6	5	**7**
Number of Sentences	10	3	12	Facebook and YouTube	**35**
Number of Performer	20	34	22	-	**10**
Professional actors	Yes	No	No	Yes	**No**
Sampling Rate	48 kHz	44.1 kHz	44.1 kHz	48 kHZ (video), 44.1kHZ (Audio)	**44.1kHZ**
Gender Balanced	Yes	Yes	No	NO	**NO**
Data Type	Audio	Audio	Audio	Video	**Audio-visual**

Table 4: Comparison Among Various Public Other Language Emotion Recognition and ANUVUTI Dataset.

Emotion Recognition Datasets in Other Languages:

Specification	RAVDESS (2018)	CREMA-D	SAVEE	TESS	CK+	EMO-DB	FER1013	ANUVUTI
Number of records (files)	7356	7442	480	2800	593	535	32298	**1035**
Number of Emotions	7	6	7	7	7	7	7	**7**
Number of Sentences	1 and song	12	12	-	-	10	-	**35**

Number of Performer	24	91	10	2	-	10	-	**10**
Professional actors	Yes	Yes	No	Yes	No	Yes	No	**No**
Sampling Rate	48 kHz	44.1 kHz	44.1 kHz	44.1k HZ	44.1k HZ	16kHz	-	**44.1KHz**
Data Type	Audio-visual	Audio-visual	Audio	Audio	video	Audio	Image	**Audio-visual**
Language	English	English	English	English	-	German	-	**Bangla**
Age (years)	-	20-74	27-31	26-64	18-50	-	-	**18-25**
Size	24.8 GB	473 MB (only audio available)	113MB	449 MB	2 MB	40MB	101MB	**497MB**

8.2.Data collection challenges:

Since emotions can differ significantly between people, countries, and situations, it's critical to record a broad range of real emotional reactions in order to develop models that generalize well to diverse scenarios. By prioritizing the realism of emotion expression in video data collection represent genuine and spontaneous expressions in a realistic way. Studies show that a specific language represents more acceptable performance on model training. To develop a separate dataset on Audio-visual for Bangla language to mitigate the lack of useful linguistics resources for emotion recognition. Several obstacles may arise when gathering video footage to create an emotion recognition dataset. For building a realistic database, we choose non-professional actors because Professional speakers sometimes overact their emotions when recording. Some actors were local Bangla participants and some were from East West University's students. Selection and recordings were done in five phases. In each phase, we collected video recordings from 2

participants using smart phones. All the phases are done a month to record from all the participants. Video data collection for emotion detection raises privacy concerns because it may reveal personal information about individuals. For this reason, we prepare concerns because People need to know exactly how their data will be used, who will have access to it, and what security will be in place to keep it private. All the participants filled up all the personal information and signed the concern form to record their video data. Some participants may find it difficult to express certain emotions openly or convincingly act out certain emotions, such as fear or sadness. Participants were given the opportunity to take sufficient time to prepare themselves to express the intended emotions authentically and for this, the environment was set up to ensure their comfort and ease. Certain participants were allowed to deliver their speeches in their regional language to accurately convey the intended expressions.

8.3.Data preprocessing:

For raw image:

To achieve optimal performance and reduce noisy or unclear data, we use different feature extraction methods. To extract an image from video data we use OpenCV library and speech from video data we use Moviepy library. Preprocessing includes framing feature detections and removing the non-speech portion for processing of image and speech data.

Normalization: Image data is optimized for emotion detection models through normalization, a process that includes scaling pixel values to a range of [0, 1]. Improving training model performance and cutting down on computational complexity. For which facilitates efficient feature learning and raises the model's emotional recognition accuracy, the input is standardized in this preprocessing stage.

Rescaling: The rescaling technique contains multiple steps to normalize the size of detected facial regions in input images. Before extracting the relevant face portions from the original photos, the bounding boxes cover the forehead region which makes subsequent analytic tasks such as emotion identification or facial expression classification easier.

Resizing: We resize the images to dimensions of 128x128 pixels using bilinear interpolation, ensuring uniformity across all facial images. This scaling procedure allows for consistent feature extraction and comparison across multiple facial photos, facilitating subsequent analysis.

Noise Reduction: We used Gaussian blur technique to remove noise. This technique is commonly used in image preprocessing to improve the quality and clarity of images. It iterates through the images in the input folder, reading each one with OpenCV and smoothing pixel values with Gaussian blur and a 5x5 kernel.

Extracted AUs: AU represents a distinct facial movement, such as brow rise or lip stretch. This approach allows for the definition of face expressions and actions, which is important for tasks like emotion recognition and facial behavior analysis.

Grayscaling: The cv2.cvtColor() method from OpenCV is used to convert each image to grayscale format. Load each image individually from the selected input folder. This conversion simplifies subsequent image processing tasks since grayscale images contain intensity values representing pixel brightness rather than discrete color channels, making them easier to analyze and manipulate for tasks like feature extraction and pattern recognition.

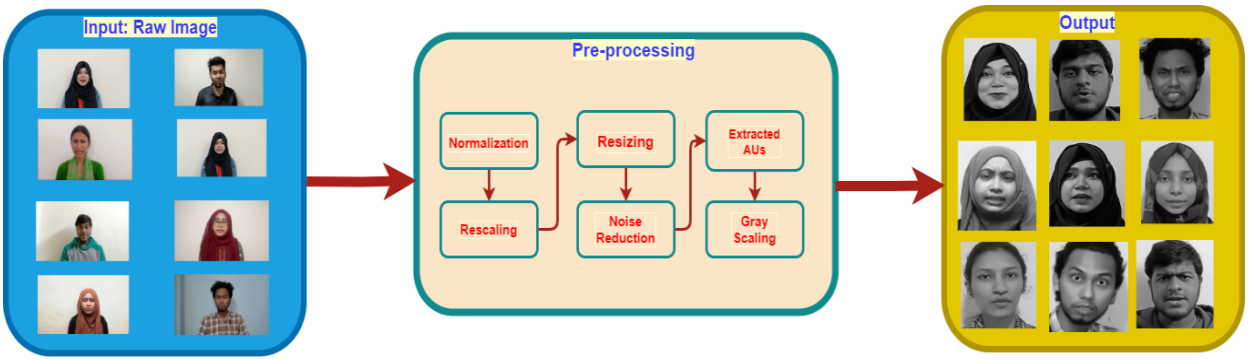

Fig 02: Data preprocessing steps for raw image.

For Extracted Speech Data:

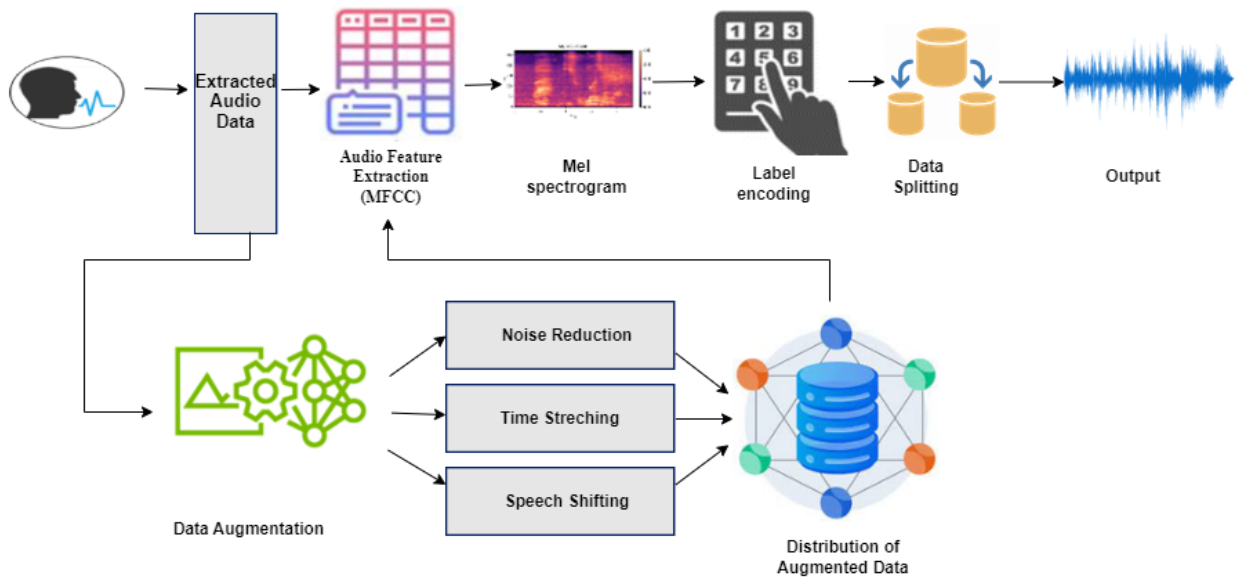

Fig 03: Data preprocessing steps for speech data.

Feature Extraction: Librosa to load audio files and extracts features called Mel-frequency cepstral coefficients (MFCCs). MFCCs are frequently employed to depict a sound's short-term power spectrum in speech and audio processing. One feature vector per audio file is obtained by computing the mean of the MFCCs.

Data Augmentation: Data augmentation is a method that involves transforming pre-existing data samples in order to artificially increase their size and diversity. Augmentation techniques for audio data can involve time-stretching the audio signals, introducing random noise, and changing the pitch or speed.

Label Encoding: The emotion labels in augmented_labels are iterated over using a list comprehension, where each label is replaced with the associated numerical value from the label_mapping dictionary. This procedure makes it easier to use the categorical emotion labels in machine learning models by converting them into numerical representations.

Data Splitting: The train_test_split function from scikit-learn is used to divide the data into training and testing sets. It separates the training and testing subsets, allocating 80% of the augmented features and the matching one-hot encoded emotion labels to training and 20% to

testing. This separation makes it possible to train the model on one set of data and evaluate its performance on a different set.

8.4.Model Training and Implementation

Two deep learning models LSTM, and custom CNN were used to train the audio dataset, and three deep learning models custom CNN, VGG16, and InceptionV3 were used to train the image dataset.

Audio Dataset Training:

- **LSTM:** We used LSTM in our audio train dataset. We used 9 additional layers to make the model better. Besides, we used the 'adam' optimizer, and 'categorical_crossentropy' loss function in the model. Here are our LSTM model layers:

Layer (type)	Output Shape	Number of Parameters
lstm	(None, 256)	264192
dropout	(None, 256)	0
dense	(None, 128)	32896
dropout_1	(None, 128)	0
dense_1	(None, 64)	8256
dropout_2	(None, 64)	0
dense_2	(None, 32)	2080
dropout_3	(None, 32)	0
dense_3	(None, 7)	231

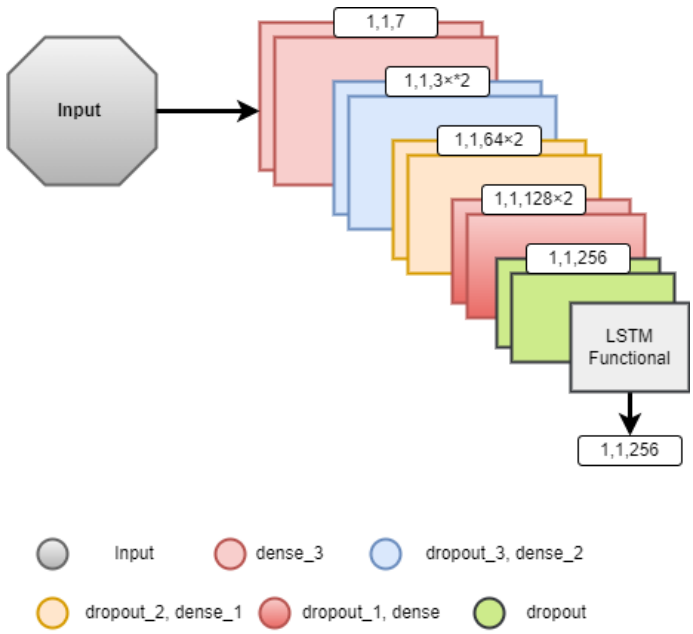

Fig 4: LSTM Layers

- **Customs CNN:** We also used the custom CNN model in our audio training dataset. We added 6 additional layers to make the model as generalized as possible to create a better model. We used the 'adam' optimizer, and 'categorical_crossentropy' loss function in the model. Our Cov2D model layers are shown below:

Layer (type)	**Output Shape**	**Number of Parameters**
Conv2d	(None, 40, 1, 64)	640
dropout	(None, 40, 1, 64)	0
flatten	(None, 2560)	0
dense	(None, 512)	1311232
dropout_1	(None, 512)	0
dense_1	(None, 7)	3591

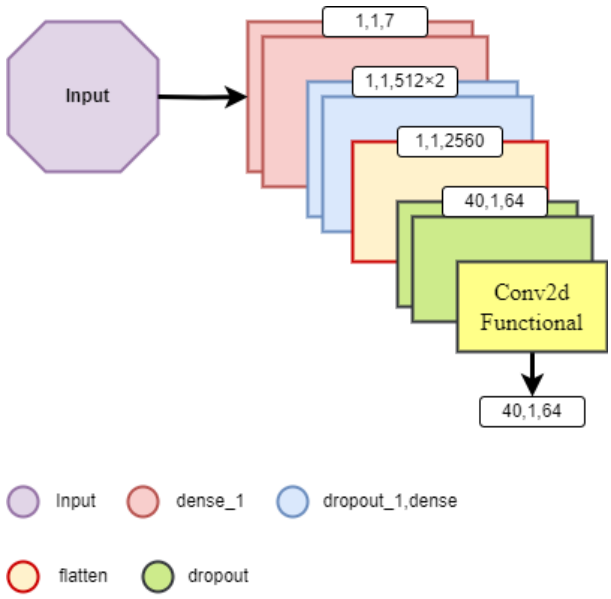

Fig 5: Customs CNN Layers(speech)

Image Dataset Training:

- **Custom CNN:** We used 14 layers of custom CNN model on our image training dataset. Here we used the 'adam' optimizer and 'categorical_crossentropy' loss function in the model. The layers that we used in our custom CNN model are shown below:

Layer (type)	**Output Shape**	**Number of Parameters**
Conv2d	(None, 62, 62, 32)	320
Conv2d_1	(None, 60, 60, 64)	18496
max_pooling2d	(None, 30, 30, 64)	0
dropout	(None, 30, 30, 64)	0
Conv2d_2	(None, 28, 28, 128)	73856
max_pooling2d_1	(None, 14, 14, 128)	0
dropout_1	(None, 14, 14, 128)	0
Conv2d_3	(None, 12, 12, 256)	295168
max_pooling2d_2	(None, 6, 6, 256)	0
dropout_2	(None, 6, 6, 256)	0
flatten	(None, 9216)	0
dense	(None, 512)	4719104
dropout_3	(None, 512)	0
dense_1	(None, 7)	3591

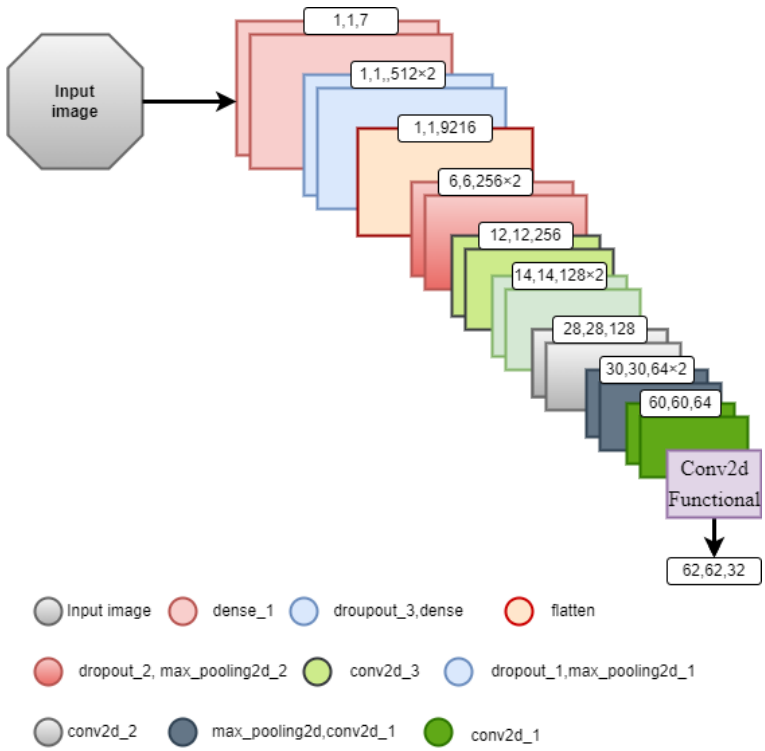

Fig 6: Customs CNN Layers(image)

- **VGG16:** We also used 8 additional layers of VGG16 model in our image training dataset. Like the previous model, the 'adam' optimizer, and 'categorical_crossentropy' loss function are also used in this model. The layers of our VGG16 model are shown below:

Layer (type)	**Output Shape**	**Number of Parameters**
vgg16 (Functional)	(None, 2, 2, 512)	14714688
flatten	(None, 2048)	0
dropout	(None, 2048)	0
dense	(None, 4096)	8392704
dropout_1	(None, 4096)	0
dense_1	(None, 1024)	4195328
dropout_2	(None, 1024)	0
dense_2	(None, 7)	7175

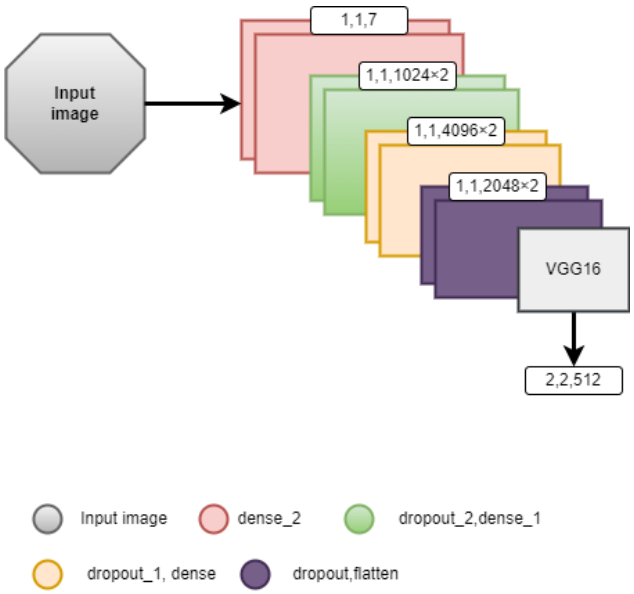

Fig 7: Vgg16 Layers

- **InceptionV3:** We also implemented the InceptionV3 model in our dataset. The model consists of approximately 93 layers, including convolutional layers, batch normalization layers, activation functions, pooling layers, concatenation layers, and fully connected layers.

9. Model Evaluation and Result Analysis:

9.1.Audio Dataset:

The experimental results will be displayed for the augmented audio dataset. Here are the results of the LSTM and custom CNN models that we used in our audio dataset.

Table 5: Result Analysis

	Model	Training Accuracy	Validation Accuracy	Testing Accuracy	Training Loss	Validation Loss
1	LSTM	99.52%	88.86%	84.74%	0.0210	0.5532
2	Custom CNN	98.76%	94.19%	94.19%	0.0396	0.3649

between those two models, the custom CNN model gives better testing accuracy. LSTM models give 84.74% testing accuracy whereas custom CNN gives 94.19%. However, we can not tell that, our deep learning models are well suited to our dataset by only observing the accuracy. One of the main things we need to do is check if the models are overfitted or not. To measure that, we run our models 100 epochs and, the training accuracy, validation accuracy, training loss, and testing loss are represented below graph:

LSTM

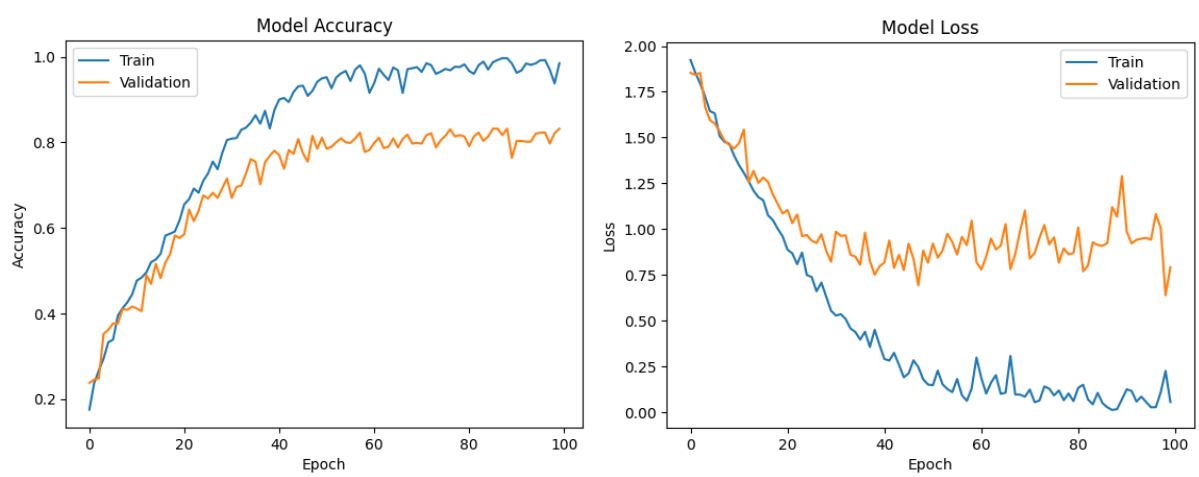

Fig: Training and Validation Accuracy/Loss of LSTM model.

Based on the graphs, it is evident that data was not distributed properly in the LSTM model. The validation accuracy should increase as much as the training accuracy. But here, we can't see it, which means the model is overfitting. The gap between validation and training accuracy is decreasing, which indicates that the model is generalizing to new data. The training loss decreases with the number of increasing epochs. The validation loss does not decrease as much as the training loss and after 100 epochs validation loss is still near 55%. Like the validation and training accuracy, the difference between training and validation loss is also decreasing with the increasing epochs.

Custom CNN

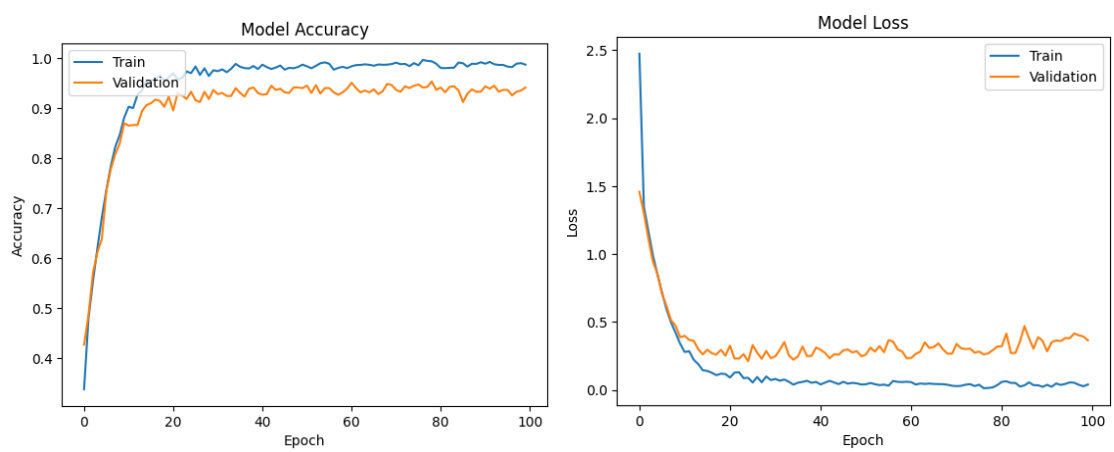

Fig: Training and Validation Accuracy/Loss of custom CNN model.

In this graph, both the training accuracy and validation accuracy increase as the number of epochs increases, which is a good sign. This suggests that the model is learning from the training data and improving its ability to classify examples correctly.

Only validation and testing accuracy are not enough to determine how well a model performs. So, other parameters such as precision, f1-score, and recall are necessary to assess how well a model performed for a particular dataset. Here are the necessary notations for calculating these results.

$$Precision = true\ positive\ /\ (true\ positive + false\ positive)$$

$$Recall = true\ positive\ /\ (true\ positive + false\ negative)$$

$$\text{F1}\ Score = 2 \times (precision \times recall)\ /\ (presion + recall)$$

LSTM model Confusion Matrix

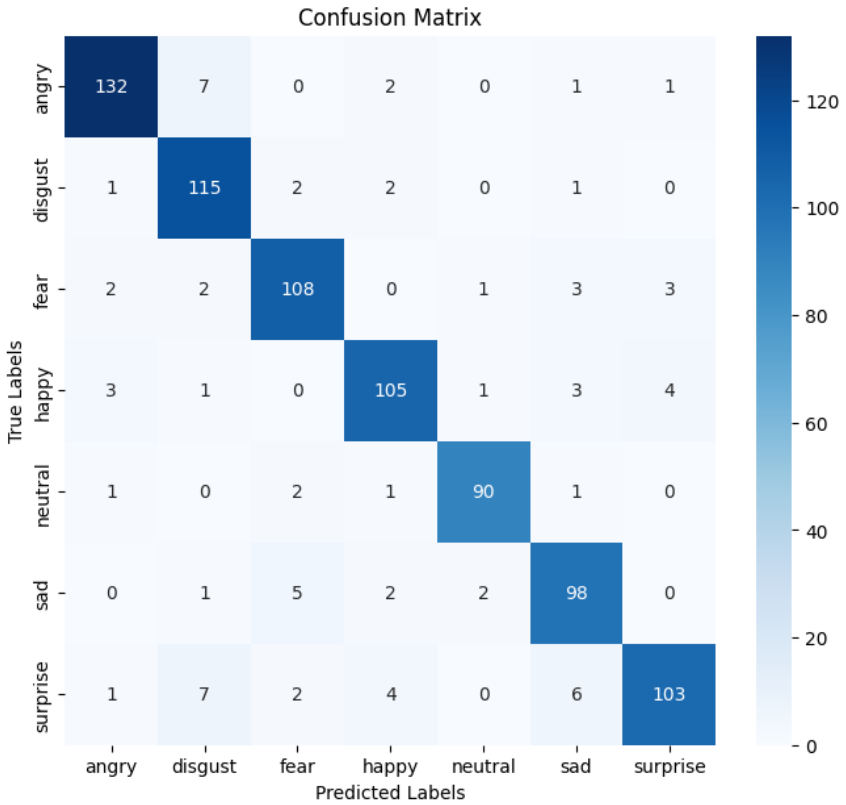

Table 6 : Precision, recall, f1-score of LSTM model

Class (for 7 classes)	Precision	recall	F1-score
angry	0.94	0.92	0.93
disgust	0.86	0.95	0.91
fear	0.91	0.91	0.91
happy	0.91	0.90	0.90
neutral	0.96	0.95	0.95
sad	0.87	0.91	0.89
surprise	0.93	0.84	0.88

Custom CNN model Confusion Matrix

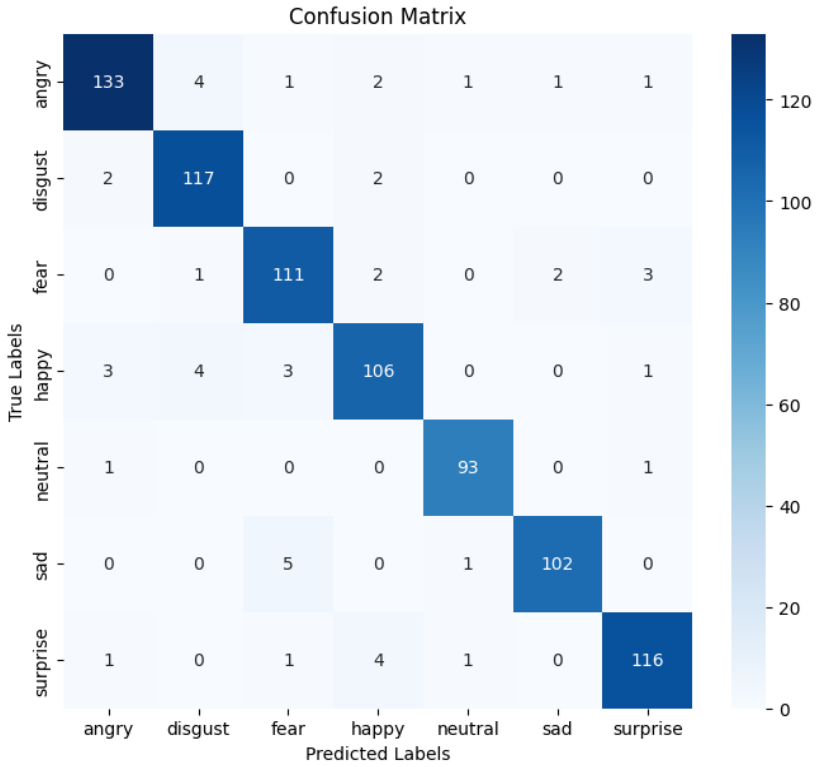

Table 7 : Precision, recall, f1-score of custom CNN model

Class (for 7 classes)	Precision	recall	F1-score
angry	0.95	0.94	0.94
disgust	0.93	0.95	0.95
fear	0.92	0.93	0.93
happy	0.91	0.91	0.91
neutral	0.97	0.97	0.97
sad	0.97	0.94	0.96
surprise	0.95	0.94	0.95

From the confusion matrix the precision, recall, and f1-score were calculated for those 7 classes. We can see both models work well to classify the test dataset. However, the custom CNN model works better than the LSTM model. So, we can sum up that, between those two models, the custom CNN model performs well in our audio dataset.

9.2.Image Dataset:

Our image dataset is already big enough to implement a deep learning model. So we didn't augment our image dataset. We applied several image models to our image dataset. Custom CNN. VGG16 and InceptionV3 models were applied to the dataset and their results are shown below:

Table 8: Result Analysis

	Model	Training Accuracy	Validation Accuracy	Testing Accuracy	Training Loss	Validation Loss
1	CNN	92.34%	96.58%	95.31%	0.2075	0.0923
2	VGG16	89.71%	85.12%	31.25%	1.0007	2.5316
3	InceptionV3	65.73%	39.02%	37.43%	0.8673	2.2996

The performance metrics of three different neural network models on a classification task over our image dataset are presented. The Convolutional Neural Network (CNN) exhibited the highest accuracy across all metrics, with a training accuracy of 92.34%, validation accuracy of 96.58%, and testing accuracy of 95.31%. It also demonstrated the lowest training and validation losses at 0.2075 and 0.0923, respectively. In contrast, the VGG16 model displayed a lower performance, especially evident in its validation and testing accuracies, which were substantially lower than its training accuracy. The InceptionV3 model showed the poorest performance, with notably lower training, validation, and testing accuracies compared to the other models, along with relatively higher losses. These results indicate that while accuracy is an essential metric for evaluating model performance, it's crucial to consider other factors like loss. Additionally, the significant gaps between training and validation/testing accuracies suggest potential overfitting in the VGG16 and InceptionV3 models.

Custom CNN

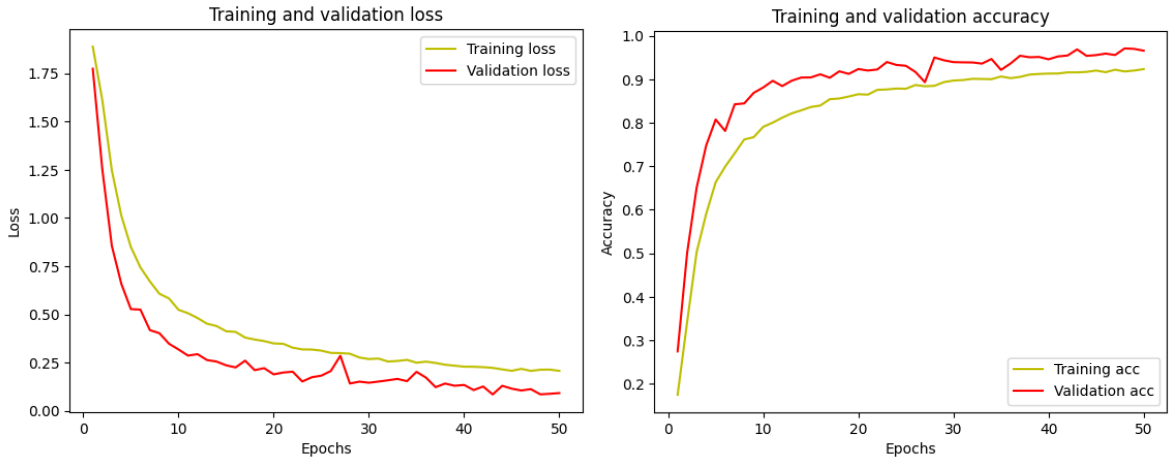

Fig: Training and Validation Accuracy of custom CNN model.

Analyzing the accuracy trends, the model seems to be learning from the training data as indicated by the increasing training accuracy. From the graph, we can say, the model is well fitted into our dataset as the training and validation loss continuously decreases and the training and validation accuracy increases over time.

VGG16

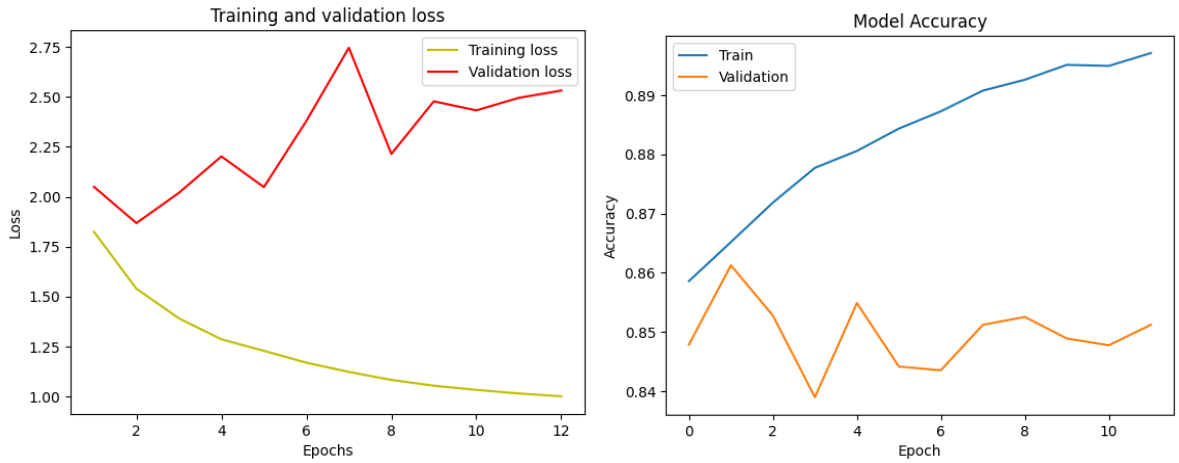

Fig: Training and Validation Accuracy of VGG16 model.

Here we run the model for 50 epochs but we used the early stopping method with a patience value of 10. The model stopped at 12 epochs due to this which indicates that the model is overfitted and stopped. So we can say the model is not well fitted to our dataset.

InceptionV3

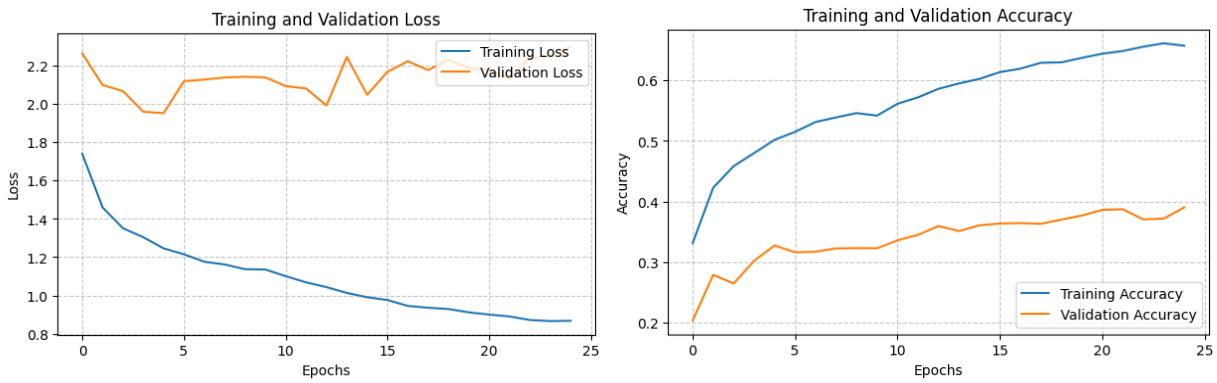

Fig: Training and Validation Accuracy of InceptionV3 model.

Here we run the model for 50 epochs but use the early stopping method with a patience value of 20. The early stopping method is used particularly in training deep learning models, to prevent overfitting and improve generalization performance. As the validation loss started increasing after the 4th epoch and in the next 20 epochs the validation loss did not improve, which indicates overfitting and the model stopped running after 24 epochs. So, we can say that the model is not well-fitted to our dataset.

By observing all those model's accuracy and loss graphs we can say our custom CNN model works better than the other two models on our image dataset. But we also need to check other necessary parameters like precision, recall, and f1-score to ensure which model is performing well on our dataset.

Here is the confusion matrix, precision, recall, and f1-score of each model:

Custom CNN model Confusion Matrix

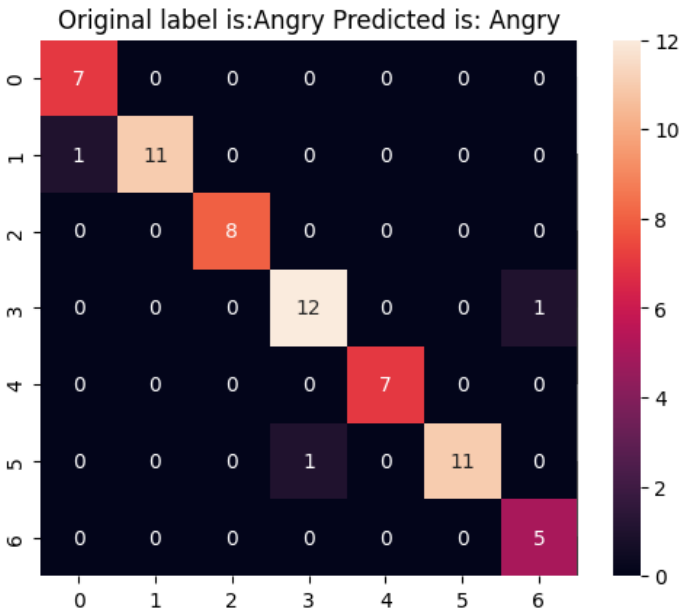

Table : Precision, recall, f1-score of custom CNN model

Class (for 7 classes)	Precision	recall	F1-score
angry	0.88	1.00	0.93
disgust	1.00	0.92	0.96
fear	1.00	1.00	1.00
happy	0.92	0.92	0.92
neutral	1.00	1.00	1.00
sad	1.00	0.92	0.96
surprise	0.83	1.00	0.91

Here we take a 64-size batch and implement our trained model on the batch to generate the above parameters.

VGG16 model Confusion Matrix

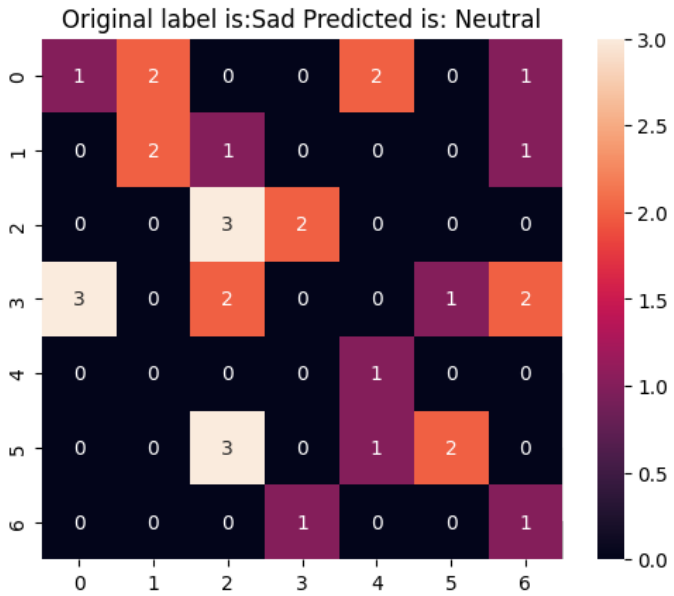

Table : Precision, recall, f1-score of custom VGG16 model

Class (for 7 classes)	Precision	recall	F1-score
angry	0.25	0.17	0.20
disgust	0.50	0.50	0.50
fear	0.33	0.60	0.43
happy	0.00	0.00	0.00
neutral	0.25	1.00	0.40
sad	0.67	0.33	0.44
surprise	0.20	0.50	0.29

Here we take a 32-size batch and implement our trained model on the batch to generate the above parameters.

InceptionV3 model Confusion Matrix

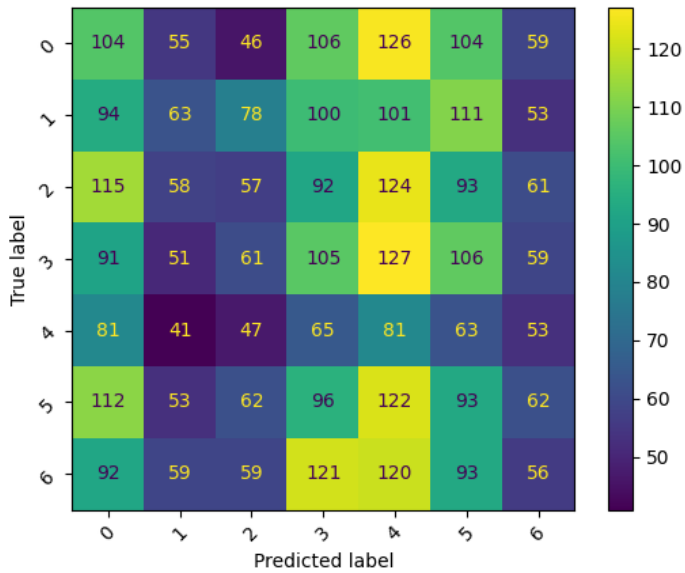

Table : Precision, recall, f1-score of custom InceptionV3 model

Class (for 7 classes)	Precision	recall	F1-score
angry	0.15	0.17	0.16
disgust	0.17	0.10	0.13
fear	0.14	0.10	0.11
happy	0.15	0.17	0.16
neutral	0.10	0.19	0.13
sad	0.14	0.15	0.15
surprise	0.14	0.11	0.11

Here we implement our trained model on the whole validation dataset to generate the above parameters.

By considering all the above analysis we can sum up that our custom CNN model works better than the VGG16 and InceptionV3 models on our dataset.

In both datasets, the custom CNN model performed better than other models. However, our task is to detect an emotion from video data. For that, we need to build a fusion model. We will use our custom CNN models for the fusion in the future.

10. Conclusion

In conclusion, this research project successfully tackled the challenge of limited emotion recognition technology for Bangla. Anuvuti, a custom-built Bangla emotion recognition dataset we created, played a vital role in training deep learning models. The custom CNN model achieved an impressive 94.19% accuracy on audio data, demonstrating the feasibility of real-time emotion recognition in Bangla speech. According to our analysis, custom CNN fared better than LSTM, exhibiting higher accuracy, precision, recall, and F1-scores in our evaluations, indicating the effectiveness of multimodal fusion.While Anuvuti represents a significant leap forward, limitations acknowledge the need for further development. Expanding the dataset and incorporating diverse participants will enhance generalizability. Exploring a broader spectrum of Bangla-specific emotions and potentially investigating more intricate models offer promising avenues for improvement.The project's true impact lies in its potential for real-world applications. Integrating Anuvuti into educational tools, customer service interfaces, and human-computer interaction systems can significantly benefit Bangla-speaking communities. Future research can leverage transfer learning from larger datasets and explore incorporating text data for a more comprehensive understanding of emotions in Bangla speech.Ethical considerations regarding user privacy, data security, and potential biases require careful attention throughout development. Transparency and responsible practices are essential to ensure this technology is used for good. Considering everything, by providing insights into multimodal fusion methods and the development of real-time emotion detection systems, which in turn encourages more organic human-computer interactions in Bangla-speaking communities.

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12. Appendix

Addressing of COs, Knowledge Profile (K), and Complex Engineering Problem (EP)

CO	K	EP
CO3	Problem Analysis [K1, K2, K3] K1: CNN used to classify images. K2: Fundamental knowledge of Artificial Intelligence, Machine Learning, deep learning, image processing, mobile app development, and web database. K3: custom CNN,LSTM ,VGG16 architecture is used.	EP3, EP6] EP1: LSTM and customs CNN models have been built and fine-tuned [K4] A hardware device has been designed and built [K5] A mobile based front end has been designed [K6] Different papers on similar goals of building CNN models to detect emotions were studied [K8]. EP2: The mobile application was optimized for efficient emotion detection based on facial expressions and speech. It was designed to be user-friendly, ensuring easy use for anyone, regardless of technical expertise. EP3: A number of models have been trained to find the model with the highest accuracy. EP6: Multiple machine learning models were developed to classify seven distinct emotions, including six basic emotions and neutral. A user-friendly interface was implemented for the mobile application.
CO4	Design and Implementation [K5] K5: Familiarity with Extensible Markup Language (XML) is essential for designing our mobile application. We utilized XML extensively in the design process of our	EP1: A solid understanding of XML is required for designing mobile applications. Additionally, knowledge of hardware and circuit design is necessary for the hardware components. EP2: We initially developed a

	mobile application.	CNN model in Python, but it wasn't efficient on Raspberry Pi. This led us to use Tiny ML to optimize performance. EP5: We used Android Studio, a modern development tool, to design and implement the mobile application. EP7: To create an emotion detection model, we first had to preprocess the dataset, then build the customs CNN and LSTM model, and finally implement the IoT device. Each of these tasks depended on the others for successful integration.
CO5	Materials and Devices [K6] K6: Knowledge about deep learning, computer vision, and digital image processing and android app development has been implemented.	Materials and Devices [EP1, EP2, EP4, EP5] EP1: Technologies like OpenCV, Keras, and TensorFlow are employed for training deep learning models, while Android Studio is utilized for developing the mobile application. EP2: Differences in results between dataset images necessitate preprocessing of every captured image. EP5: The project employs modern tools and adheres to standard design and development processes.